

Combined and Conditional Probability

AND means multiply, OR means add. Without replacement: one fewer each time. “Given that” shrinks the group you divide by.



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THE TOPIC ON ONE MAP

Combined and Conditional Probability

01 – AND and OR

× for and + for or at least one

02 – Without replacement

two picks three picks list the orders

03 – Conditional probability

“given that” “if ... then”

04 – Algebra in probability

form an equation quadratic

THE FOUR WORDS Read the question for these words first:

ZONE AND
both happen
multiply

ZONE OR
one or the other
add the ways

ZONE AT LEAST ONE
one, two or all
 $1 - P(\text{none})$

ZONE GIVEN THAT
we already know
divide by that group only

01 AND and OR: Combining Independent Events

Independent means one result does not change the other. Then $P(A \text{ and } B) = P(A) \times P(B)$. For OR, add the separate ways — or use $P(A) + P(B) - P(A \text{ and } B)$ when both can happen.

WHEN YOU SEE → YOU DO

“both”, “and”

→ multiply

“either ... or ... or both”

→ $P(A) + P(B) - P(A)P(B)$

“at least one of the four dice”

→ $1 - P(\text{none})$

“the same colour” (with replacement)

→ add $P(RR) + P(GG) + \dots$

“what assumption has Alfie made?”

→ that the events are independent

WORKED MODEL • JANUARY 2020 P1HR

$P(\text{win tennis}) = 0.6$, $P(\text{win both}) = 0.42$. Independent. $P(\text{wins neither})$?

1 $0.6 \times P(\text{chess}) = 0.42 \Rightarrow P(\text{chess}) = 0.7$

AND = ×

2 lose tennis 0.4, lose chess 0.3

1 - P

ANSWER $0.4 \times 0.3 = 0.12$

× **TRAP** Adding for “both”. Two things both happening is **less** likely than either one: multiply.

IDEA 01

Practise: AND and OR: Combining Independent Events

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM AND / OR

CH 6 • TOPIC 6 2 marks

$P(A) = 0.3$, $P(B) = 0.5$, independent. Find $P(A \text{ and } B)$ and $P(A \text{ or } B)$.

SELECTED PRACTICE

Q2**PRACTICE**

Practice • Medium

CH 6 • TOPIC 6 3 marks

- 1 (a)** When a drawing pin is dropped it can land point down or point up.

Lucy, Mel and Tom each dropped the drawing pin a number of times.

The table shows the number of times the drawing pin landed point down and the number of times the drawing pin landed point up for each person.

	Lucy	Mel	Tom
point down	31	53	16
point up	14	27	9

Rachael is going to drop the drawing pin once.

Whose results will give the best estimate for the probability that the drawing pin will land point up?

Give a reason for your answer.

(1 mark)

- (b)** Stuart is going to drop the drawing pin twice.

Use all the results in the table to work out an estimate for the probability that the drawing pin will land point up the first time and point down the second time.

(2 marks)

Q3**PRACTICE**

Practice • Medium

CH 6 • TOPIC 6 2 marks

13 A bag contains 20 discs.

10 are red, 7 are blue and 3 are green.

Marnie takes a disc at random before putting it back in the bag. Nick then takes a disc at random before putting it back in the bag.

Olly then takes a disc at random.

Work out the probability that they all take a red disc.

(2 marks)**Q4****PRACTICE**

Practice • Hard

CH 6 • TOPIC 6 3 marks

3 Thelma spins a biased coin twice. The probability that it will come down heads both times is 0.09

Calculate the probability that it will come down tails both times.

(3 marks)

Q5**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 6**4 marks**

11 Liam is trying to remember a 3-digit code. He knows the rule that

the first digit is a cube number

the second digit is a factor of 16

the third digit is an odd number.

Liam tries at random a code that matches the rule.

Work out the probability that this is the correct code.

(4 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 1HR • Q16**CH 6 • TOPIC 6** **4 marks**

16 Steffi is going to play one game of tennis and one game of chess.

The probability that she will win the game of tennis is 0.6

The probability that she will win **both** games is 0.42

Work out the probability that she will **not** win either game.

.....

(Total for Question 16 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q16

CH 6 • TOPIC 6 3 marks

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

(Total for Question 16 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q14

CH 6 • TOPIC 6 3 marks

14 Ciara throws **four** fair six-sided dice.

The faces of each dice are labelled with the numbers 1, 2, 3, 4, 5, 6

Work out the probability that at least one of the dice lands on an even number.

(Total for Question 14 is 3 marks)

02 Without Replacement: Two or Three Picks

After each pick the total drops by one, and so does the count of that colour. Write each pick as a fraction and multiply. If the order does not matter, count every order.

WHEN YOU SEE → YOU DO

“takes two / three at random”

→ without replacement — totals $n, n - 1, n - 2$

“one of each colour”

→ RB and BR: two orders

“at least one of each colour” (3 picks)

→ $1 - P(\text{all red}) - P(\text{all yellow})$

“still in the bag”

→ think about what was taken out

Fraction for each pick

Multiply

Count the orders

Add

WORKED MODEL • MAY 2024 P1H

15 red and 5 yellow sweets; Fred takes 3. Find $P(\text{at least one of each colour})$.

$$1 \quad P(RRR) = \frac{15}{20} \cdot \frac{14}{19} \cdot \frac{13}{18} = \frac{2730}{6840}$$

all red

$$2 \quad P(YYY) = \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18} = \frac{60}{6840}$$

all yellow

ANSWER $1 - \frac{2790}{6840} = \frac{45}{76}$

× **TRAP** Forgetting the second order. “One red and one blue” is RB **or** BR.

IDEA 02

Practise: Without Replacement: Two or Three Picks

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Three picks

CH 6 • TOPIC 6 3 marks

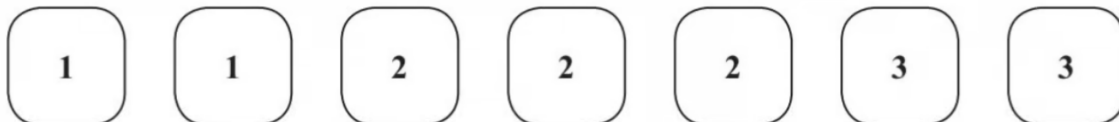
A bag has 3 red and 5 blue counters. Three are taken without replacement. Find $P(\text{all blue})$.

SELECTED PRACTICE

Q10**PRACTICE**

Practice • Medium

CH 6 • TOPIC 6 5 marks

4 (a) Here are seven tiles.

Jim takes at random a tile.
He does **not** replace the tile.

Jim then takes at random a second tile.

Calculate the probability that both the tiles Jim takes have the number 1 on them.

(2 marks)

(b) Calculate the probability that the number on the second tile Jim takes is greater than the number on the first tile he takes.

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 6 • TOPIC 6 5 marks

- 15** Dani has a pack of 45 cards.
Each card is either red or black.

One-third of the cards in the pack are **red**.

She picks two cards from the pack, without replacement.

Calculate the probability that Dani picks two **black** cards.

(5 marks)

Q12**PRACTICE**

Practice • Hard

CH 6 • TOPIC 6 4 marks

8 (a) There are 12 counters in a bag.

There is an equal number of red counters, blue counters and yellow counters in the bag.
There are no other counters in the bag.

3 counters are taken at random from the bag.

Work out the probability of taking 3 red counters.

(2 marks)

(b) The 3 counters are put back into the bag.

Some more counters are now put into the bag.

There is still an equal number of red counters, blue counters and yellow counters in the bag.
There are no counters of any other colour in the bag.

3 counters are taken at random from the bag.

Is it now less likely or equally likely or more likely that the 3 counters will be red?

You must show how you get your answer.

(2 marks)

Q13**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 6 3 marks

13 A bag contains 30 discs.

10 are red and 20 are blue.

One disc is taken out at random and replaced by **two** of the other colour.

Another disc is then taken out at random and replaced by **two** of the other colour.

Another disc is then taken out at random.

Work out the probability that all three discs taken out are **red**.

(3 marks)

PAST PAPERS

Q14 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2H • Q20**CH 6 • TOPIC 6** **3 marks**

20 There are 12 counters in a bag.

- 3 of the counters are red
- 9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

.....

(Total for Question 20 is 3 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q16

CH 6 • TOPIC 6 4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

(Total for Question 16 is 4 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q22

CH 6 • TOPIC 6 3 marks

22 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

03 Conditional Probability: “Given That”

“Given that” tells you something has already happened, so the whole group becomes only those cases. The bottom of the fraction is that smaller group.

WHEN YOU SEE → YOU DO

“given that the student plays football”

→ divide by the footballers only

“if a bus is old, it starts first time with 0.3”

→ that number sits on the branch after “old”

“two chosen from those who used a train”

→ start from the train group, no replacement

“show that $P(\dots) = P(\dots)$ ”

→ work both sides out fully, then compare

all 40 students

football 25

swim 10

$$P(\text{swim} | \text{football}) = \frac{10}{25} = \frac{2}{5}$$

the bottom is 25, not 40

WORKED MODEL • A BUS COMPANY

25% of buses are old. Old: starts first time 0.3. Newer: 0.65.
 $P(\text{starts first time})$?

1 old path: $0.25 \times 0.3 = 0.075$

branch after “old”

2 new path: $0.75 \times 0.65 = 0.4875$

branch after “new”

ANSWER $0.075 + 0.4875 = 0.5625$

IDEA 03

Practise: Conditional Probability: “Given That”

My questions first, then selected practice.

Q17 **DR ESLAM** Given that

CH 6 • TOPIC 6 2 marks

Of 40 students, 25 play football; 10 of those also swim. A footballer is chosen. Find $P(\text{swims})$.

SELECTED PRACTICE**Q18** **PRACTICE** Practice • Medium

CH 6 • TOPIC 6 3 marks

- 11** Sophie takes an examination.
If she fails the examination, she will resit.

The probability that Sophie passes the examination on her first attempt is 0.7

If she fails the examination on any attempt, the probability she passes on the next attempt is 0.9

Work out the probability that Sophie takes at most 2 attempts to pass the examination.

(3 marks)

Q19**PRACTICE**

Practice • Medium

CH 6 • TOPIC 6 4 marks

14 A bus company has a large number of buses. 25% of the buses are more than 10 years old.

If a bus is more than 10 years old, the probability that it will start first time is 0.3.

If a bus is less than 10 years old, the probability that it will start first time is 0.65.

Amir is asked to drive one of the company's buses, chosen at random.

Calculate the probability that the bus starts first time.

(4 marks)

Q20**PRACTICE**

Practice • Hard

CH 6 • TOPIC 6 6 marks

14 21 people travelled to a meeting.

- 12 used a train.
- 6 used a car.
- 7 did not use a train or a car.
- Some used a train and a car.

Two people are chosen at random from those who used a train.

Find the probability that both these people also used a car.

(6 marks)

Q21 PRACTICE Practice • Very Hard

CH 6 • TOPIC 6 8 marks

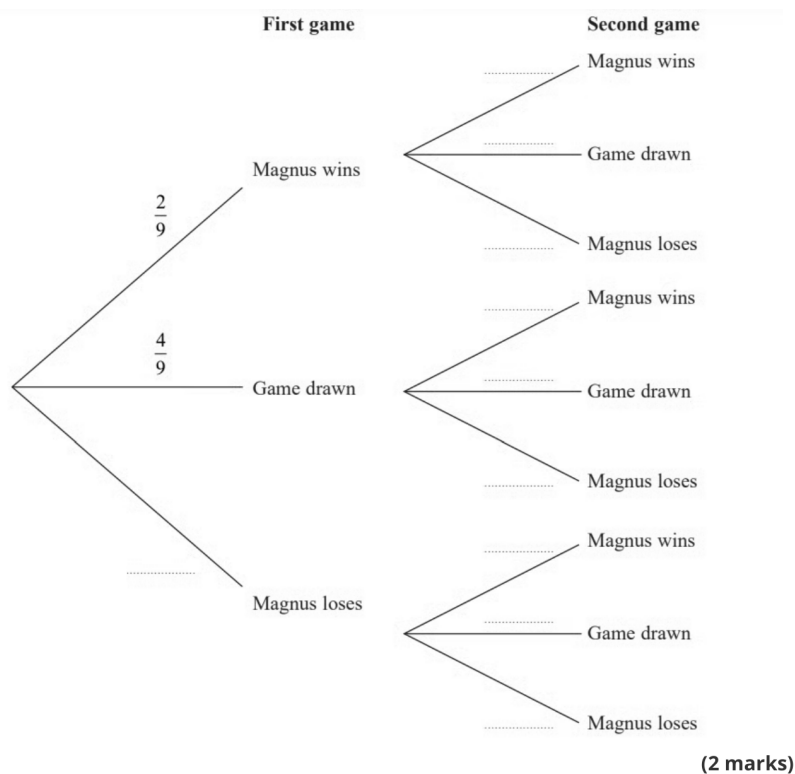
6 (a) Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

Complete the probability tree diagram.



(b) For each game of chess,

the winner gets 2 points and the loser gets 0 points,

when the game is drawn, each player gets 1 point.

Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3 marks)

(c) Magnus and Garry now play a third game of chess.

Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(3 marks)

Q22**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 6**4 marks**

- 14** Bag X contains 9 blue balls and 18 red balls.
Bag Y contains 7 blue balls and 14 red balls.

Liz picks a ball at random from bag X.
She puts the ball into bag Y.
Mike now picks a ball at random from bag Y.

Show that

$$P(\text{Liz picks a blue ball}) = P(\text{Mike picks a blue ball})$$

(4 marks)

04 Algebra in Probability

When a count is unknown, call it x (or n), write each probability with it, multiply along the path, and set it equal to the given probability. Clear the fractions — you usually get a quadratic.

WHEN YOU SEE → YOU DO

“there are n sweets, 6 are orange”

$$\rightarrow \frac{6}{n} \cdot \frac{5}{n-1}$$

“show that $n^2 - n - 90 = 0$ ”

→ cross-multiply and collect on one side

“find the possible values of N ”

→ keep both positive whole-number roots

a table with x and y

→ add to 1, plus the extra equation: simultaneous

Name the unknown

Write the path

Equal to the given

Solve

WORKED MODEL • n SWEETS

n sweets, 6 orange. Two eaten. $P(\text{both orange}) = \frac{1}{3}$. Find n .

$$1 \quad \frac{6}{n} \cdot \frac{5}{n-1} = \frac{1}{3} \Rightarrow 90 = n(n-1)$$

cross-multiply

$$2 \quad n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0$$

quadratic

ANSWER $n = 10$

× **TRAP** Keeping the negative or fraction root. A number of sweets is a positive whole number.

IDEA 04

Practise: Algebra in Probability

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Form an equation

CH 6 • TOPIC 6 4 marks

A bag has n counters, 4 of them red. Two are taken without replacement.
 $P(\text{both red}) = \frac{2}{15}$. Find n .

SELECTED PRACTICE**Q24** PRACTICE Practice • Hard

CH 6 • TOPIC 6 5 marks

9 There are 10 pens in a box.

There are x red pens in the box.

All the other pens are blue.

Jack takes at random two pens from the box.

Find an expression, in terms of x , for the probability that Jack takes one pen of each colour.

Give your answer in its simplest form.

(5 marks)

Q25**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 6 7 marks

4 (a) There are y black socks and 5 white socks in a drawer.

Joshua takes at random two socks from the drawer.

The probability that Joshua takes one white sock and one black sock is $\frac{6}{11}$

Show that $3y^2 - 28y + 60 = 0$

(4 marks)

(b) Find the probability that Joshua takes two black socks.

(3 marks)

Q26**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 6 5 marks

9 Pippa has a box containing N pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

(5 marks)

Q27**PRACTICE**

Practice · Very Hard

CH 6 · TOPIC 6 5 marks

- 16** There are 7 red counters in a bag.
The rest of the counters in the bag are blue.

There are more blue counters than red counters in the bag.

Two counters are to be taken at random from the bag.

The probability that there will be one counter of each colour is $\frac{7}{15}$

Work out the total number of counters in the bag before any counters are taken from the bag.

(5 marks)

PAST PAPERS

Q28**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q12

CH 6 • TOPIC 6 5 marks**12** Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	x	0.15	0.5	y	0.13	0.03

(a) Show that $x + y = 0.19$ Given that $3x - y = 0.09$ and $x + y = 0.19$

(b) work out the value of x and the value of y
Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 12 is 5 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | AND and OR | AND $\times$, OR $+$; at least one = $1 - \text{none}$. |
| 02 | No replacement | One fewer each pick; count every order. |
| 03 | Given that | Divide by the smaller group only. |
| 04 | Algebra | Unknown $\rightarrow$ path = given $\rightarrow$ quadratic. |

BOTTOM LINE AND $\times$ • OR $+$ • no replacement $\rightarrow$ one fewer • given $\rightarrow$ smaller bottom.

Answers

Combined and Conditional Probability • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • AND AND OR: COMBINING INDEPENDENT EVENTS

- Q1** Dr Eslam • AND / OR $0.15; 0.3 + 0.5 - 0.15 = 0.65$
- Q2** Practice • Medium (a) Mel: the most drops (80) (b) $P(\text{up}) = \frac{50}{150} = \frac{1}{3}; \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$
- Q3** Practice • Medium $\frac{10}{20} \times \frac{10}{20} \times \frac{10}{20} = \frac{1}{8}$
- Q4** Practice • Hard $P(H) = \sqrt{0.09} = 0.3$, so $P(T) = 0.7: 0.7^2 = 0.49$
- Q5** Practice • Very Hard first digit 1 or 8, second 1, 2, 4, 8, third 1, 3, 5, 7, 9: $\frac{1}{2} \times \frac{1}{4} \times \frac{1}{5} = \frac{1}{40}$
- Q6** Edexcel 4MA1 • January 2020 • Paper 1HR • Q16 0.12
- Q7** Edexcel 4MA1 • November 2021 • Paper 2H • Q16 $\frac{77}{225}$
- Q8** Edexcel 4MA1 • June 2022 • Paper 2HR • Q14 $1 - \left(\frac{1}{2}\right)^4 = \frac{15}{16}$

IDEA 02 • WITHOUT REPLACEMENT: TWO OR THREE PICKS

- Q9** Dr Eslam • Three picks $\frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} = \frac{5}{28}$
- Q10** Practice • Medium (a) $\frac{2}{7} \times \frac{1}{6} = \frac{1}{21}$ (b) $\frac{2}{7} \cdot \frac{5}{6} + \frac{3}{7} \cdot \frac{2}{6} = \frac{16}{42} = \frac{8}{21}$
- Q11** Practice • Medium 15 red, 30 black: $\frac{30}{45} \times \frac{29}{44} = \frac{29}{66}$
- Q12** Practice • Hard (a) $\frac{4}{12} \cdot \frac{3}{11} \cdot \frac{2}{10} = \frac{1}{55}$ (b) e.g. 5 of each: $\frac{5}{15} \cdot \frac{4}{14} \cdot \frac{3}{13} = \frac{2}{91} > \frac{1}{55}$ — more likely
- Q13** Practice • Very Hard $\frac{10}{30} \times \frac{9}{31} \times \frac{8}{32} = \frac{3}{124}$
- Q14** Edexcel 4MA1 • June 2023 • Paper 2H • Q20 $\frac{219}{220}$
- Q15** Edexcel 4MA1 • May 2024 • Paper 1H • Q16 $\frac{45}{76}$
- Q16** Edexcel 4MA1 • November 2025 • Paper 1H • Q22 $\frac{34}{35}$

IDEA 03 • CONDITIONAL PROBABILITY: “GIVEN THAT”

- Q17** Dr Eslam • Given that $\frac{10}{25} = \frac{2}{5}$
- Q18** Practice • Medium $0.7 + 0.3 \times 0.9 = 0.97$
- Q19** Practice • Medium $0.25 \times 0.3 + 0.75 \times 0.65 = 0.5625$
- Q20** Practice • Hard train and car = $12 + 6 - 14 = 4$: $\frac{4}{12} \times \frac{3}{11} = \frac{1}{11}$
- Q21** Practice • Very Hard (a) every set of branches: $\frac{2}{9}, \frac{4}{9}, \frac{3}{9}$ (b) $\frac{28}{81}$ (c) $\frac{208}{729}$
- Q22** Practice • Very Hard Liz: $\frac{9}{27} = \frac{1}{3}$; Mike: $\frac{1}{3} \cdot \frac{8}{22} + \frac{2}{3} \cdot \frac{7}{22} = \frac{22}{66} = \frac{1}{3}$

IDEA 04 • ALGEBRA IN PROBABILITY

- Q23** Dr Eslam • Form an equation $\frac{4}{n} \cdot \frac{3}{n-1} = \frac{2}{15} \Rightarrow n^2 - n - 90 = 0 \Rightarrow n = 10$
- Q24** Practice • Hard $2 \times \frac{x}{10} \times \frac{10-x}{9} = \frac{x(10-x)}{45}$
- Q25** Practice • Very Hard (a) $\frac{10y}{(y+5)(y+4)} = \frac{6}{11} \Rightarrow 3y^2 - 28y + 60 = 0$ (b) $y = 6$:
 $\frac{6}{11} \times \frac{5}{10} = \frac{3}{11}$
- Q26** Practice • Very Hard red r , black $r + 3$: $r^2 - 15r + 54 = 0 \Rightarrow r = 6$ or 9 : $N = 15$ or $N = 21$
- Q27** Practice • Very Hard blue b : $b^2 - 17b + 42 = 0 \Rightarrow b = 14$ (more than 7): 21 counters
- Q28** Edexcel 4MA1 • May 2023 • Paper 1HR • Q12 $x = 0.07, y = 0.12$